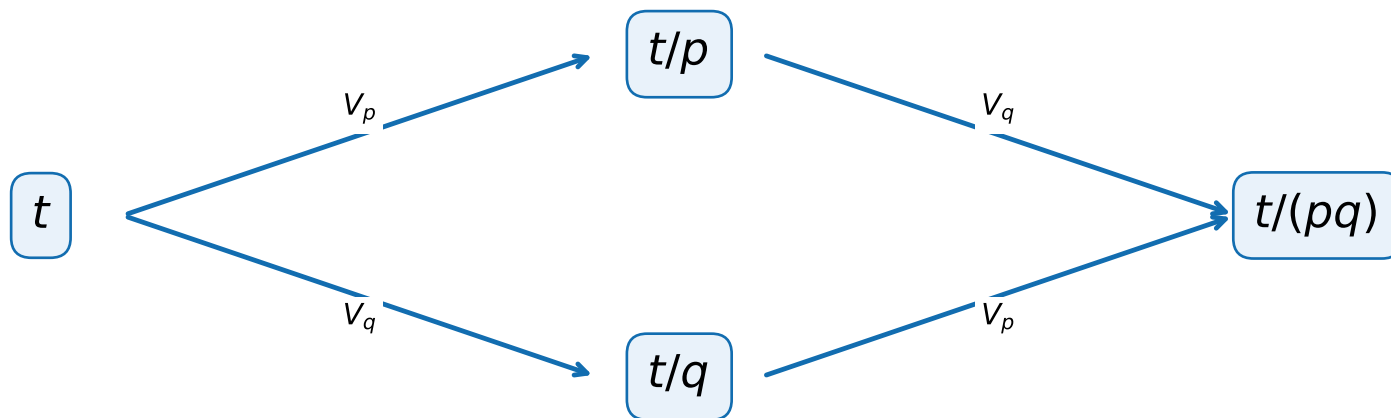


Prime factors act on the whole shift family



$$V_p V_q = V_{pq} = V_q V_p, \quad (V_p f)(t) = f(t/p)$$

$$\zeta(s, 1+t) = \left[\prod_p (1 - p^{-s} V_p)^{-1} \right] (1+t)^{-s}, \quad \text{Re } s > 1$$

At $t = 0$: every prime fixes the parameter; scalar Euler factors are recovered.

Full support: $\Phi_R(r, \lambda) = (r, 1_K \otimes \lambda)$ retains every $\lambda \in L$.

Exact parameter diagram; it does not identify the parameter fixed point with a theorem about all zeros.
 Proof: HURWITZ_PRIME_SCALING.tex. Human sources: Bost-Connes; Connes-Consani-Marcolli, Fun with F_1 .